

Practice Quiz 07A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

State the epsilon-delta definition and use it once.

1. (a) Let $f : \mathbf{R} \rightarrow \mathbf{R}$ and let $c, L \in \mathbf{R}$. State, with all quantifiers, the definition of $\lim_{x \rightarrow c} f(x) = L$.

Solution

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in \mathbf{R} (0 < |x - c| < \delta \rightarrow |f(x) - L| < \varepsilon). \quad \blacksquare$$

- (b) For $f(x) = 4x$, prove from the definition that $\lim_{x \rightarrow 1} f(x) = 4$.

Solution Let $\varepsilon > 0$ be arbitrary and choose $\delta = \varepsilon/4 > 0$. Let $x \in \mathbf{R}$ be arbitrary and suppose $0 < |x - 1| < \delta$. Then

$$|f(x) - 4| = |4x - 4| = 4|x - 1| < 4\delta = \varepsilon.$$

This verifies the definition, so $\lim_{x \rightarrow 1} 4x = 4$. $\blacksquare$